

Akshyasree Ravi

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EDUCATION:

Rensselaer Polytechnic Institute, (RPI), Troy, NY

May 2028

Bachelor of Science in Industrial & Management Engineering, Minor in Cognitive & Behavioral Neuroscience

Software: Excel, SQL, Power BI, Minitab, Siemens NX (CAD), MATLAB, Python, PowerPoint, Canva

Certifications/Licenses: Six Sigma Green Belt for Healthcare, AHA Basic Life Support Certificate

Relevant Courses: Deterministic Methods in OR, Human Performance Modeling & Support, Production & Operations Management, Modeling & Analysis of Uncertainty, Intro to Engineering Design, Information & Data Systems

Languages: Proficient Tamil Speaker & Reader, Proficient Korean Reader, Beginner Korean Understanding/Speaking

PROJECT/RESEARCH EXPERIENCE:

Engineering Design Project Member – Portable Air-Curtain Dust Extraction System, RPI

Summer 2026

- Co-designed a portable air-curtain extraction system to reduce airborne particulate exposure during woodworking and construction operations
- Performed engineering calculations for airflow rate, nozzle sizing, exit velocity, plenum geometry, baffle placement, and pressure distribution to optimize contaminant capture efficiency
- Conducted **workflow analysis** and **user-centered risk assessment** to improve operator safety and usability
- Designed CAD models in **Siemens NX** and developed 3D-printed prototypes, balancing manufacturability, performance, and ergonomic constraints
- Built, tested, and iteratively refined **physical prototypes** using airflow testing and experimental observations to validate and improve system performance

Paid Undergraduate Research Assistant – Strength of Materials Simulation Project (URP), RPI

Aug 2025-Dec 2025

- Developed interactive engineering simulations supporting undergraduate Strength of Materials instruction.
- Built computational models using MATLAB and Python to simulate structural loading and material behavior
- Collaborated with faculty to improve visualization tools and educational interfaces for engineering concepts
- Integrated simulation outputs with course materials to improve student engagement and engineering education

Regeneron Honors Science Research Program, Yorktown Heights, NY

Sep 2021-June 2024

- Designed and conducted a three-year behavioral science research study evaluating the implementation of Dialectical Behavior Therapy (DBT) skills in secondary education
- Developed and implemented the "**Husker Discovery**" curriculum, which continues to be used at Yorktown High School
- Performed statistical analysis using hypothesis testing (t-tests) to evaluate intervention outcomes
- Presented research findings at regional science competitions and technical conferences
 - Awarded Third Place** at 2024 WESEF, 2024 Tri-County Science Fair, and 2023 Somers Science Fair

LEADERSHIP:

Undergraduate Teaching Assistant – Engineering Communications, RPI

Aug 2026-Present

- Support weekly class sessions by guiding students through Siemens NX design work, technical drawings, and Excel tools
- Grade assignments, provide feedback, and attend professor meetings to support course instruction

U.S. Marines Platoon Leadership Course Program, Albany, NY

May 2026-Present

- Planning, leading, and running Physical Training sessions for 10-20 PLC program participants in the Capital District area
- Recruit and mentor prospective applicants through campus outreach
- Develop leadership, decision-making, and team coordination skills in a structured officer development program

Director of External Communications - Institute of Industrial & Systems Engineers (IISE), RPI

Aug 2024-Present

- Develop and execute communication strategy for the RPI IISE chapter through digital outreach and event promotion
- Design marketing materials for technical workshops, professional development events, and recruiting initiatives
- Manage Instagram account (@iise.rpi), develop campaigns that increase student engagement and event participation
- Appointed Safety Officer** - ensuring proper Environment, Health, and Safety for the student chapter

RPI Representative - 2026 IISE Annual National Conference & Expo, Arlington, TX

May 2026

- Attended workshops and panels featuring industry speakers, research presentations, and professional networking

RPI U.S. Army ROTC Program Cadet, Troy, NY

Aug 2024-May 2026

- Served as Squad Leader:** responsible for accountability, mentorship, and daily coordination of assigned 5+ cadets
- Completed leadership development through field training, operational planning, and team-based exercises

VOLUNTEER EXPERIENCE:

English Teacher/Ophthalmologist Volunteer - Tiruchirappalli Thillainagar Lions Club, Tamil Nadu, India

Summer 2025

- Assisted healthcare professionals in providing vision screenings to **over 1,000** underserved students
- Taught introductory English communication and grammar to **over 100** elementary school students
- Contributed to community improvement and environmental service initiatives